



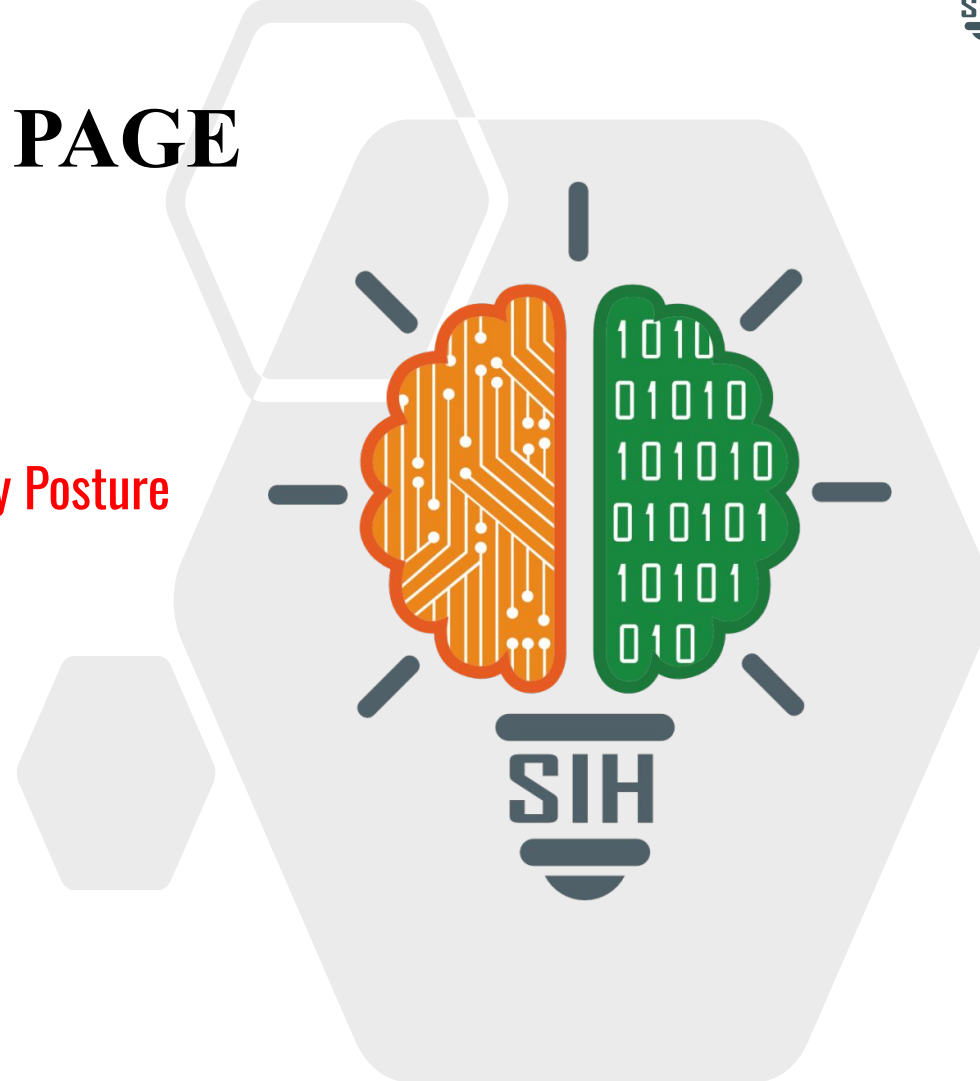
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TITLE PAGE

- **Problem Statement ID – 26159**
- **Problem Statement Title-**
SecureMailScope : AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name: Cryterpillars**



Cryterpillars
SMALL PACKETS. BIGGER SECURITY.
ANALYZE | DETECT | SECURE | EVOLVE

AI-Assisted Cryptographic Security Posture Assessment

Problem

- Email infrastructure can have **TLS enabled but still remain vulnerable**
- Hidden weaknesses include:
 - Deprecated TLS versions
 - Weak cipher suites
 - Expired / mismatched certificates
 - STARTTLS downgrade
 - Cleartext authentication
 - Lack of forward secrecy
- Existing packet-analysis tools primarily expose **raw network evidence**, leaving correlation and risk assessment to analysts.

Our Solution

PCAP/PCAPNG → Security Intelligence

1. Reconstruct SMTP / IMAP / POP3 sessions
2. Detect STARTTLS/STLS transitions
3. Analyze TLS versions & cipher suites
4. Extract & validate X.509 certificates
5. Detect cryptographic weaknesses
6. AI-assisted risk & anomaly detection
7. Generate **Security Posture Score**
8. Provide prioritized remediation

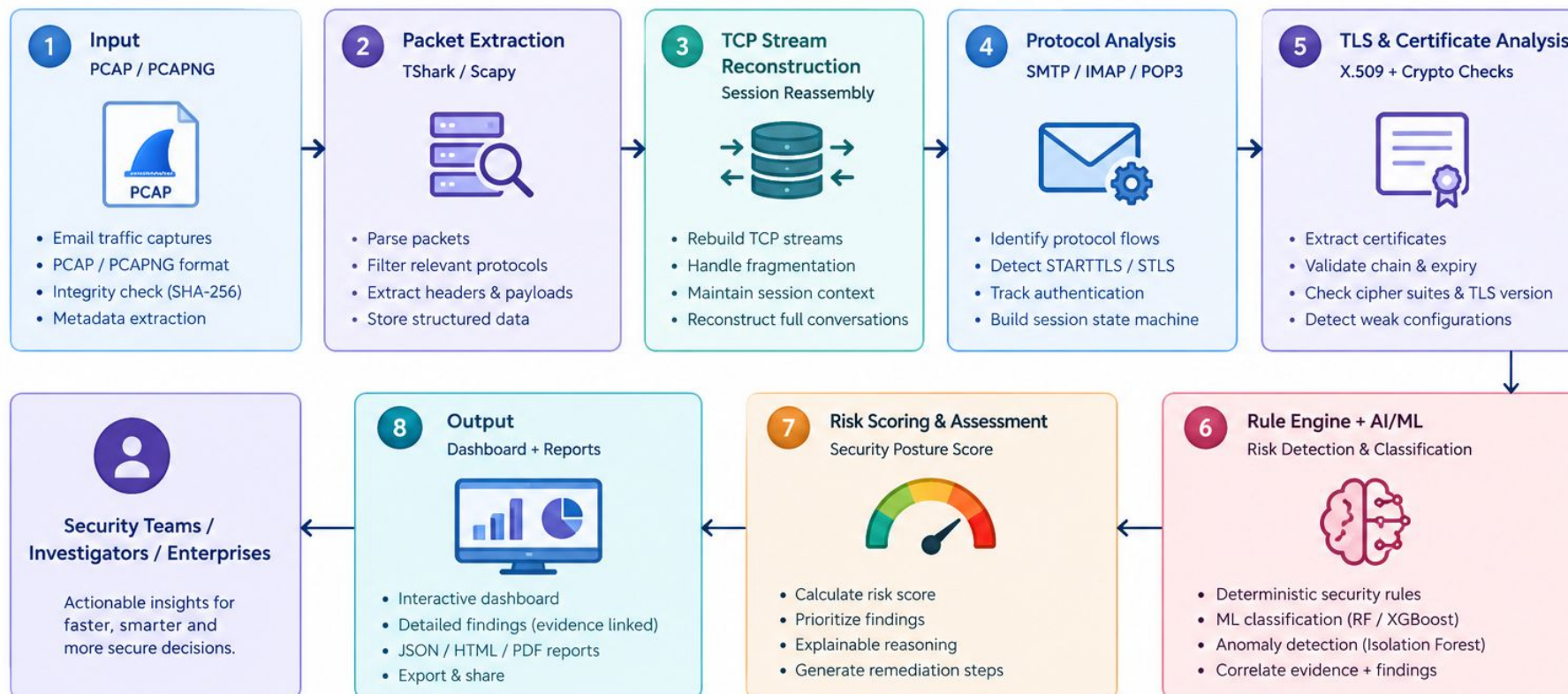
Key Innovation

From Packet Inspection → Explainable Cryptographic Reasoning → Evidence → Finding → Risk → Remediation

The proposed platform is designed as a passive forensic system that converts email traffic captures into an explainable security posture assessment.

TECHNICAL APPROACH

System Architecture



Tech Stack

Backend: Python, FastAPI

Packet Analysis: TShark, Scapy, PyShark

TLS/X.509: Python cryptography, OpenSSL

AI/ML: Random Forest / XGBoost + Isolation Forest

Database: PostgreSQL / SQLite

Frontend: React + TypeScript + Tailwind + Plotly

Deployment: Docker Compose, Offline-first

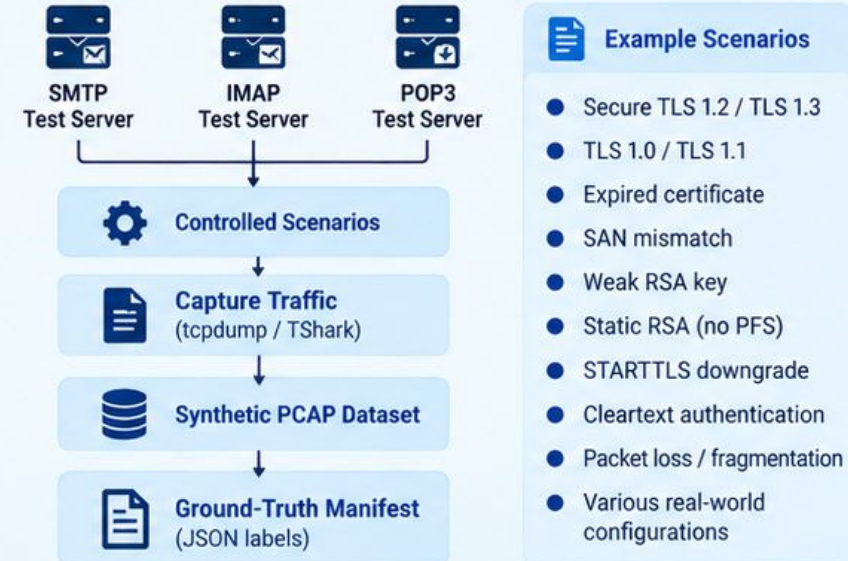
AI Approach

Deterministic Security Rules + ML Classification + Anomaly Detection
AI operates on **normalized cryptographic features**, not arbitrary raw packet text.

✓ Why It Is Feasible?

- ✓ **Mature Open-Source Tools**
Uses TShark, Scapy, OpenSSL, etc.
- ✓ **Standard PCAP/PCAPNG Support**
Works with widely available formats
- ✓ **Modular Architecture**
Easy to develop, extend and maintain
- ✓ **Rule-Based Detection**
Provides deterministic and reliable results
- ✓ **ML Layer for Enhanced Detection**
Can be trained using controlled synthetic traffic
- ✓ **Offline Deployment Possible**
No sensitive data leaves the environment

Dataset Generation Pipeline



⚠ Challenges and Mitigation

Challenge	Mitigation Strategy
Incomplete PCAP files	Evidence quality classification, mark confidence levels
Encrypted payloads	Analyze TLS metadata and handshake instead of content
False AI predictions	Deterministic rule validation + human review
Sensitive email data	Offline/local processing, no third-party APIs
TCP fragmentation / packet loss	Sequence-aware stream reconstruction
Certificate revocation (OCSP/CRL)	Explicit offline validation status
Diverse real-world configurations	Build synthetic + real-world test scenarios
Scalability for large PCAPs	Modular pipeline with parallel processing

🚀 Deployment Strategy



📈 Overall Viability





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IMPACT AND BENEFITS



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For SOC Teams

Detect. Triage. Respond Faster.

Less Time
on Manual Work
More Time on Security

- ✓ Faster incident triage
- ✓ Reduced manual packet analysis
- ✓ Prioritized security findings
- ✓ Repeatable cryptographic audits



For Forensic Investigators

From Packets to Proven Evidence.

Find the
Full Story

- ✓ Evidence-preserving analysis
- ✓ Packet/frame-linked findings
- ✓ Timeline reconstruction
- ✓ Exportable forensic reports



For Enterprises & Government

Stronger Security. Greater Trust.

Secure
Institutions
Stronger Nation

- ✓ Certificate health monitoring
- ✓ TLS configuration assessment
- ✓ Configuration-drift detection
- ✓ Compliance-oriented security posture
- ✓ Offline deployment for sensitive environments



Key Benefits

A SIMPLE PATH
TO A SAFER
EMAIL ECOSYSTEM



RAW PCAP

Network packet
capture files



Automated Security Analysis

AI-powered cryptographic
and configuration analysis



Explainable Risk Score

Clear, evidence-backed
severity scoring



Prioritized Remediation

Actionable recommendations
for a stronger security posture

Broader Benefits

BEYOND SECURITY
TOWARDS A SAFER
DIGITAL TOMORROW



Social Impact

- ✓ More secure digital communication for citizens
- ✓ Protection against phishing and email-based attacks
- ✓ Builds trust in government and public services



Economic Benefits

- ✓ Reduces cost of security breaches
- ✓ Lower operational effort for security teams
- ✓ Improves organizational resilience and productivity



Environmental Benefits

- ✓ Enables local/offline analysis, reducing data transfer
- ✓ Minimizes need for repeated hardware audits
- ✓ Supports sustainable and secure digital infrastructure

Standards & Protocols

- **NIST SP 800-52 Rev. 2** — TLS Security Guidelines — <https://csrc.nist.gov/pubs/sp/800/52/r2/final>
- **RFC 3207** — SMTP STARTTLS — <https://www.rfc-editor.org/rfc/rfc3207.html>
- **RFC 8314** — Secure Email Communication — <https://www.rfc-editor.org/rfc/rfc8314.html>

Technical References

- **Wireshark / TShark** — PCAP & Protocol Analysis — <https://www.wireshark.org/docs/>
- **Scapy** — Network Packet Analysis — <https://scapy.readthedocs.io/>
- **Python Cryptography** — X.509 & Certificate Validation — <https://cryptography.io/>
- **OpenSSL** — TLS & Cryptographic Operations — <https://docs.openssl.org/>

AI / ML

- **Scikit-learn** — Random Forest & Isolation Forest — <https://scikit-learn.org/>
- **XGBoost** — Risk Classification — <https://xgboost.readthedocs.io/>

Dataset & Evaluation

- Synthetic SMTP / IMAP / POP3 PCAP Dataset
- Ground-Truth Security Labels
- Precision, Recall & F1-Score
- Precision@K for Anomaly Detection
- Byte-Level Accuracy for Stream Reconstruction